GCCF - Assignment #3

SECTION B (2*4=8)

Q1. (CO3) What is VM?

Ans - A virtual machine (VM) is a digital version of a physical computer. Virtual machine software can run programs and operating systems, store data, connect to networks, and do other computing functions, and requires maintenance such as updates and system monitoring. It's essentially a software-based emulation of a physical computer system. VMs allow you to run multiple operating systems (OS) or instances of the same OS on a single physical machine.

Q2. (CO3) What is Load Balancer?

Ans- A Load Balancer is a device or software component that distributes incoming network traffic across multiple servers or resources in a balanced manner. Its purpose is to optimize resource utilization, maximize throughput, minimize response time, and avoid overloading any single resource. Load balancers are commonly used in high-traffic websites, web applications, and server clusters to ensure scalability, reliability, and availability.

Q3. (CO3) Differentiate between IPV4 and IPV6.

Ans- Difference Between IPv4 and IPv6

IPv4	IPv6
IPv4 has a 32-bit address length	IPv6 has a 128-bit address length
It Supports Manual and DHCP address configuration	It supports Auto and renumbering address configuration
In IPv4 end to end, connection integrity is Unachievable	In IPv6 end-to-end, connection integrity is Achievable

IPv4	IPv6
It can generate 4.29×109 address space	The address space of IPv6 is quite large it can produce 3.4×1038 address space
The Security feature is dependent on the application	IPSEC is an inbuilt security feature in the IPv6 protocol
Address representation of IPv4 is in decimal	Address Representation of IPv6 is in hexadecimal
Fragmentation performed by Sender and forwarding routers	In IPv6 fragmentation is performed only by the sender
In IPv4 Packet flow identification is not available	In IPv6 packet flow identification are Available and uses the flow label field in the header
In IPv4 checksum field is available	In IPv6 checksum field is not available
It has a broadcast Message Transmission Scheme	In IPv6 multicast and anycast message transmission scheme is available
In IPv4 Encryption and Authentication facility not provided	In IPv6 Encryption and Authentication are provided
IPv4 has a header of 20-60 bytes.	IPv6 has a header of 40 bytes fixed
IPv4 can be converted to IPv6	Not all IPv6 can be converted to IPv4

IPv4	IPv6
IPv4 consists of 4 fields which are separated by addresses dot (.)	IPv6 consists of 8 fields, which are separated by a colon (:)
IPv4's IP addresses are divided into five different classes. Class A , Class B, Class C, Class D , Class E.	IPv6 does not have any classes of the IP address.
IPv4 supports VLSM(Variable Length subnet mask).	IPv6 does not support VLSM.
Example of IPv4: 66.94.29.13	Example of IPv6: 2001:0000:3238:DFE1:0063:0000:0000:FEFB

Q4. (CO3) Explain instance.

Ans- An instance refers to a single occurrence or example of something, particularly in the context of computing. In computing, an instance typically refers to a specific realization of a software component, such as a program, process, or virtual machine, running on a computer system.

SECTION C (6*2=12 Mark)

Q1. (CO3) Write an example of simple pipeline.

A simple pipeline example could involve processing and analyzing text data. Here's a basic outline:

1. Data Collection: Gather text data from various sources such as documents, websites, or databases.

2. Preprocessing:

- Text Cleaning: Remove any irrelevant characters, symbols, or formatting.
- Tokenization: Split the text into individual words or tokens.
- Lowercasing: Convert all text to lowercase to ensure consistency.
- Stopword Removal: Eliminate common words (e.g., "the", "and") that don't add significant meaning.

3. Feature Extraction:

- Bag-of-Words (BoW): Represent each document as a vector containing word frequencies.
- TF-IDF (Term Frequency-Inverse Document Frequency): Assign weights to words based on their importance in the document and across the corpus.

4. Model Training:

- Choose a machine learning model such as a classifier (e.g., Naive Bayes, Support Vector Machine) or clustering algorithm (e.g., K-means).
- Split the data into training and testing sets.
- Train the model on the training data.

5. Evaluation:

-Assess the performance of the model using metrics such as accuracy, precision, recall, or F1-score.
- Adjust parameters or try different models if necessary.

6. Inference:

- Apply the trained model to new, unseen text data to make predictions or extract insights.

This pipeline illustrates a typical workflow for text data analysis, encompassing data preprocessing, feature extraction, model training, evaluation, and inference.

Change Data Capture Pipeline

This is a streaming Change Data Capture (CDC) pipeline that captures any changes to the data at the source and sends it along to the destination. In this case, the source is a SQL server database, and the destination is two cloud warehouses: Databricks and Snowflake.

Q2. (CO3) What is REST API?

Ans- A **REST API** is a type of *application programming interface* (API) that complies with the ***representational state transfer*** (REST) model of data representation and communication between two systems (a client and server) over a network such as the Internet. REST APIs support information exchange between internal and third-party applications, and enable businesses to integrate multiple endpoints into their application ecosystem.

REST API stands for Representational State Transfer Application Programming Interface. It is an architectural style for designing networked applications, particularly web services, where communication occurs over the HTTP protocol.

Key principles of RESTful APIs include:

1. Statelessness: Each request from a client to the server must contain all the necessary information to understand and process the request. The server should not store any client state between requests.

2. Resource-based: Resources (such as data objects or representations of entities) are identified by unique URIs (Uniform Resource Identifiers). Clients interact with these resources through standard HTTP methods like GET, POST, PUT, DELETE, etc.

3. Uniform Interface: The interface between client and server should be uniform and consistent. This includes standard HTTP methods, resource URIs, and representations (such as JSON or XML) for data exchange.

4. Client-Server Architecture: The client and server are separate entities that communicate via HTTP requests and responses. This separation allows for scalability and portability.

5. Cacheability: Responses from the server can be marked as cacheable or non-cacheable, allowing clients to reuse responses to improve performance.

RESTful APIs are commonly used for building web services that enable communication and interaction between different systems or components over the internet. They provide a flexible and scalable way to expose functionality and data to clients in a platform-independent manner.

SECTION D (10*2=20Mark)

Q1. (CO3) Define customer-managed encryption keys (CMEK).

Ans- Customer-Managed Encryption Keys (CMEK) is a feature offered by cloud service providers that allows customers to retain control over the encryption keys used to encrypt their data stored in the cloud. With CMEK, customers generate, manage, and store their own encryption keys outside of the cloud provider's infrastructure.

Key aspects of CMEK include:

1. Control: Customers have full control and ownership of the encryption keys, including the ability to generate, rotate, and revoke keys as needed.

2. Security: By managing their encryption keys, customers can implement their own security policies and procedures to safeguard their data.

3. Compliance: CMEK can help customers meet regulatory and compliance requirements by providing assurances that they retain control over their data encryption.

4. Isolation: Encryption keys are stored separately from the encrypted data, providing an additional layer of security and isolation.

5. Integration: CMEK integrates with cloud services, allowing customers to seamlessly encrypt and decrypt data using their own keys while still benefiting from the scalability and flexibility of cloud infrastructure.

Overall, Customer-Managed Encryption Keys empower customers with greater control and security over their data in the cloud environment.

Q2. (CO3) Explain the abstraction layers of Tensor flow and gsutil. What was the solution if any organization has 5TB of private data on premises and they need to migrate the data to Cloud Storage with maximize the data transfer speed?

Ans- TensorFlow and gsutil operate at different levels of abstraction, but both are tools commonly used in the Google Cloud Platform (GCP) ecosystem.

1. TensorFlow Abstraction Layers:

- TensorFlow is an open-source machine learning framework developed by Google. It provides various abstraction layers for building and deploying machine learning models.

- Abstraction layers in TensorFlow include:

  - Low-level APIs: These offer fine-grained control over model building, training, and deployment. Developers can define computational graphs and execute operations directly.

  - High-level APIs: These provide easier-to-use interfaces for common machine learning tasks. Examples include tf.keras for building neural networks and tf.estimator for training and evaluation.

2. gsutil Abstraction Layers:

- gsutil is a command-line tool for interacting with Google Cloud Storage (GCS), which is a scalable object storage service offered by Google Cloud.

- Abstraction layers in gsutil include:

  - Low-level Object Operations: These allow users to perform basic operations such as copying, moving, and deleting objects in GCS buckets.

  - High-level Commands: These provide convenient shortcuts for tasks like syncing local directories with GCS buckets, setting access control permissions, and managing bucket lifecycles.

Regarding the migration of 5TB of private data from on-premises to Cloud Storage with maximized data transfer speed, several strategies can be employed:

1. Use gsutil's Parallel Copying: gsutil supports parallel copying, which can significantly speed up data transfer by utilizing multiple threads or processes. This can be achieved by specifying the `-m` flag when copying data.

2. Use Transfer Appliances: Google offers solutions like Transfer Appliance, which are physical devices that can securely transfer large volumes of data to Google Cloud Storage. This can be particularly useful for large datasets where transferring over the network would be time-consuming.

3. Optimize Network Bandwidth: Ensure that the organization's network infrastructure is optimized for data transfer. This may involve increasing available bandwidth, reducing network congestion, or prioritizing data transfer traffic.

4. Data Compression and Encryption: Compressing data before transferring it to Cloud Storage can reduce transfer times. Additionally, if encryption is required, ensure that it's performed efficiently to minimize overhead.

5. Use Cloud Storage Transfer Service: Google Cloud offers a Transfer Service specifically designed for migrating large datasets to Cloud Storage. It can handle data transfers from various sources, including on-premises environments, and provides features for optimizing transfer speeds.

By employing a combination of these strategies and leveraging the capabilities of tools like gsutil and Google Cloud's Transfer Service, organizations can effectively migrate large volumes of data to Cloud Storage while maximizing transfer speeds.